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B-LPoET: A middleware lightweight Proof-of-Elapsed Time (PoET) for efficient distributed transaction execution and security on Blockchain using multithreading technology

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ABSTRACT

A private permissioned chain is typically considered a fully protected authorized blockchain. The concept of a permissioned chain has become better single-entity control operation over the network and allows entry only to selected nodes, which most probably supports different small-medium enterprises and government officials. It is one of the prominent features of blockchain technology that maintains transaction integrity and security throughout the deliverance. This paper initially highlights the benefits of a permissioned private chain. Present the current issues of the Proof-of-Elapsed Time consensus mechanism for the distributed applicational environment. In order to make blockchain systems more adaptive for private chains, this paper proposes a middleware lightweight consensus based on PoET, named "B-LPoET". Objecting to being improved as a lightweight PoET from the predefined PoET in terms of verifying transactions and creating more blocks, to decide the mining right, and participants allows to acknowledge their identities before joining. The design of B-LPoET reduces the cost of multi-node efficiency by establishing a lightweight topology for selecting and waiting time of winning node using multithreading, which affects positively by increasing systems scalability. The experimental results of the B-LPoET illustrate that it performs better as compared with the existing PoET and other state-of-the-art consensus.

1. Introduction

A new paradigm in information and communication technology creates various new pathways for future developments, such as one of the evaluations of Blockchain Distributed Ledger Technology (BDLT) in the Industrial Internet of Things (IIoT) environment. The initial concept of Blockchain underlying that support the operations of Bitcoin is proposed by Nakamoto in 2008 [1]. For instance, a

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decentralized nature with multi-point maintenance capability handles distributed storage structures by comprehensive use of smart contracts, consensus policies, hash encryption mechanisms, intercommunication channels between nodes' interconnectivity, and consortium infrastructure. It has the characteristic of integrity, transparency, traceability, immutability, and automation [2,3]. In order to entertain different applicational domains with the use of blockchain, such as Artificial Intelligence (AI), Internet of Things (IoT), and Augmented Intelligence. It is to improve the integrity and reliability of systems in accordance with the requirement of the public chain, private chain, and protected consortium chain [3]. However, the public permissionless ledger network, including Ethereum, Solidity, and Avalanche is completely open to all nodes because of their decentralized nature. All nodes can join the chain in accordance with the chronological ordering formation, where they can access the ledger, initiate a transaction, submit a request, and leave the chain without getting any hindrance. On the other side, the private permissioned network of blockchain is only used by individuals and enterprises to protect communication among others [4]. Generally, the network is not open to outsiders, and there is a certain degree of centralized control in terms of node identity verification and validation [5,6]. Whereas consortium chain means a collaborative feature with public permissionless and private permissioned structures that make the system able to restrict write-only and read-only transactions for the organization and other participating stakeholders [5]. A multi-stream direction of consortium allows multi-companies and institutes to achieve substantial and strong integration. Thus, it can promote the ordered and successful development of the blockchain in the industry.

In the Small-medium enterprises (SMEs) environment, the role of a permissioned private network is envisioned to provide a closely protected chain environment with an access control layer infrastructure. This additional security layer only gives permission or authorization to those stakeholders who participate earlier/register to perform the transactional actions [5,6]. For Instance, a participating unit would need authorization from the expert of the SMEs network to become part of the chain. After this, the unit can read, write, and access transactional information on the blockchain. In addition, the benefit of a private chain is that it allows the creation of roles of participating stakeholders that dictate who can perform which task and contributes to the blockchain. It supports identity verification customization with the use of chain codes, which automate the validation process instead of having the network expert's permission for approval. For this reason, the permissioned private chain is considered to be partially decentralized with distributed node execution facilities. However, the critical features of the permissioned blockchain are as follows: (i) decisions are authorized by a private group, (ii) partial decentralization, (iii) transactional transparency is not required, and (iv) high privacy and security. The main purpose is to use a permissioned private blockchain network is that it provides flexibility in terms of incremental/fully centralized along with privacy protection. It increases freedom among businesses to connect without having to worry about the risks associated, such as malicious attacks, which are involved in the existing centralized network.

Throughout the features of blockchain DLT, a consensus is a critical element, which is considered a crucial part that ensures the efficient cooperation of the BDLT [6]. While reviewing the working scenario of blockchain DLT, how to create each node to keep their records continuously via a contract is a critical limitation. To achieve this, various consensus algorithms are designed, that task to provide consistency and correctness to the distributed ledger on different nodes [7]. This requires an alarming alert in terms of learning from the current mechanisms for achieving state consensus in the permissioned private environment. It determines the algorithm for choosing nodes of accounting in the distributed private network. And so, ensure correct consensus in accordance with the forms of data involved in the overall distributed chain environment. The current BDLT private consensus mechanisms are categorized into Poof-of-Work (PoW), Proof-of-Stake (PoS), Delegated Proof-of-Stake (DPoS), Proof-of-Authority (PoA), and Proof-of-Elapsed Time (PoWT) [7,8].

As one of the functions of blockchain DLT is completely decentralized in nature. The individual/large number of recorded transactions in dynamic time is very hard to handle, there may be the complexity of what is true value. The consensus algorithms are designed, that provide a single or multi-path execution of the stakeholders' requested transaction facilities; along with that, it resists malicious activities, including double spending attacks over the distributed network [8,9]. The consensus policies help in structuring to prioritize the speed, privacy, and scalability of the DLT, as shown in Fig. 1. The PoW, PoS, DPoS, and PoA consensus are mainly listed and applied in a public network environment. While certain customization of these mechanisms is applicable in the private chain,

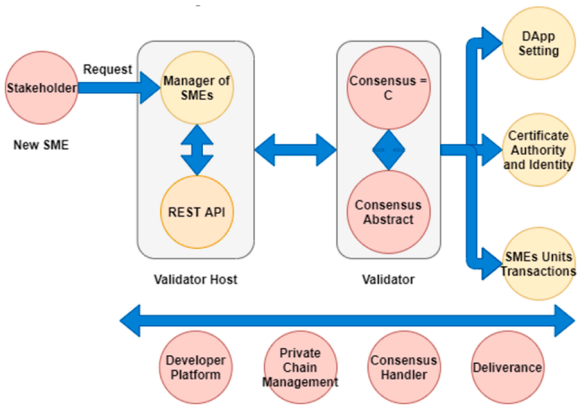

Fig. 1. The Existing Working Hierarchy of Consensus [3].

where transactions are validated by an expert of the organization. However, this consensus helps to reach/execute investing tokens on the chain, which increases the cost of services, unscheduled request proposals, and no confirmation consistency available, as shown in Fig. 1. The permissioned private blockchain mostly used Byzantine Fault Tolerance (BFT) and distributed consensus mechanism RAFT to execute the scheduled transactions. However, RAFT has gotten a variety of limitations, including downtime, node validation-related issues that lead to data forgery, and unable to handle problems of

BFT. On the other end, BFT general problem solver impossibly to get try on consistency between nodes via message passing from one to another on a single channel in the permissioned chain. It receives a large rate of message loss while managing transactions toward delivery. In order to solve the problem of BFT consensus, Practical Byzantine Fault Tolerance (PBFT) is proposed [10], where all participating nodes are connected in a sequential order with one node being the primary node and the others referred to as the backup. It improves instant transaction finality along with providing great performance in terms of less downtime it requires. At the same time, PBFT faces heavy node communication, restricted chain size, and Sybil attacks susceptible to related problems.

However, Proof-of-Elapsed Time (PoET) is considered a prominent consensus algorithm in a private permissioned chain environment. It is because PoET is suitable and applied to most of the enterprises' structures because no excessive communication is required between nodes to reach a consensus. Along with that, it provides high privacy and consumes less computational power to execute transactions. The working operation of PoET elaborates that an individual participating node in the private network requires to wait for a certain time period (randomly chosen time), where the first node to complete the desegrated wait wins the block for transaction execution [10,11]. This algorithm proposed to ensure that the participating stakeholder gets scheduled transaction execution time randomly, which is not shorter duration. In order to win the scheduled time, the winner has intended to complete the execution cycle. It is more appropriate in the IIoT environment, where each unit needs to schedule transactions of industrial, manufacturing, and production in accordance with the transactional nodes' dependency and their random time selection. In the same way, there are some drawbacks associated with this algorithm, such as relying on a trustworthiness distributed application (DApp) platform, random and limited block time, a small subset for the limited organizational environment, and only accessible in the private chain.

1.1. Motivation and objectives

This paper initially discusses the existing problem raised in the private permissioned blockchain consensus and related executions. It includes more cost of transaction executions, high computation power consumption, network bandwidth for deliverance, scalability between nodes, and scheduling/offloading complexity. And so, explain the importance of permissioned blockchain in the SMEs that robust businesses executions without creating potential risk throughout in terms of privacy protection. The protected chronological structure of the private distributed network with NuCypher Threshold *Re*-Encryption (NTRE) and identity verification mechanism increases transactional integrity throughout the delivery. The role of the participating stakeholders is assigned while joining the chain (such as reading, writing, and accessing). For instance, the advantage of using PoET in the SME environment for the purpose of secure and smooth transaction exchanges between units is discussed. However, after analysis of the complexity involved in the current PoET and the purpose of the proposed "B-LPoET", a middleware lightweight PoET for efficient distributed transaction execution and security with multithreading technology. It mainly reduces the reliance on the trustworthiness DApp, decreases block time, and variable size of blocks, a proper set of organizational environment develops, and makes them accessible in the consortium chain.

1.2. Contributions of this paper

The major contributions achieved by this paper are highlighted as follows:

- This paper addresses a list of existing research gaps involving the blockchain private permissioned consensus that require experts more consideration.
- The focus of this paper is to propose a middleware lightweight PoET consensus of permissioned private blockchains for futuristic SMEs.
- Smart contracts are created and deployed for the purpose to verify and validate requests of a new enterprise, participating in stakeholder transaction initiation lists, adding, and updating new transactional logs and deliverance records.
- At the end of this paper, a few interoperable DApp design, implementation, and deployment-related challenges and limitations are addressed.

1.3. Statement of section distribution

The subdivision of this paper is structured and presented as follows: The next Section (2) highlights the research gaps involving the existing blockchain DLT, nodes connectivity, distinct chain management, consensus algorithms, and related privacy preservation. The systematic investigation process that extracts the current problem, description of the problem, fundamental knowledge requires, and procedure to achieve this paper's objectives are discussed in Section 3. In Section 4, the proposed lightweight PoET is presented along with their working operations and related executions. The evaluation metrics/experimental results of the proposed work with the comparison of other state-of-the-art is discussed in Section 5. Finally, this paper concludes in Section 6.

2. Related works and cases

Scalability and security are the concerning factors in the existing public, private, and consortium blockchain solutions. For this reason, blockchain distributed ledger technology (BDLT) eliminates the need for trust from any third-party entities. In fact, it controls data flows automatically by deploying chain codes associated with DLT execution protocols [12]. This is how blockchain allows only authorized candidates in the network that probably handles each transaction (with minimum loss function) to occur throughout. And so, manages and validate requests in accordance with the types of data entries. The distributed nature of blockchain technology requires a consensus algorithm to schedule and organize transactional concerns of participating stakeholders and exchanges [13]. Proof-of-Elapsed Time (PoET) is one of the consensus solutions (proposed by Intel Corporation) considering in private chain, which follows Nakamoto-style consensus executions based on the time-lottery method. A random generated elapsed time decides which node of blocks gets executed first; along with that, it enhances integrity and transparency by enabling lottery results among the explicit participants. In the overall process of PoET execution, the system consumes more computational energy and other related resources.

The key concept of this research is to examine the need of industrial, manufacturing, and production units to reach a consensus about the time of integration and development depending on the collaborative decision. By analyzing mathematically, the 'I' number of industrial units and 'K' number of products are scheduled; so that, the consensus reaches time to execute most probably, such as $I = 4K + 1$. It indicates that ¼ time is required to win the transactional block time and 2/4 slice for their execution is expected. In this regard, it is most crucial to secure the network against malicious, such as connection interruption, device failure, and process failures during events of transaction executions. Therefore, various features highlighted that need more consideration for an efficient distributed

Table 1
Blockchain Permissioned Private Consensus Related Works.

Research Technique/ Methodology	Year of Study	Research Objectives	Challenges	Similarity and Differences
A novel multithreading proof-based method in blockchain consensus mechanism [14]	2022	The detail information about each consensus protocol operations Provide new protocols, comprehensive security features, and comparative analysis, such as lightweight proof-of-work, proof-of-stake, etc. is presented	Lack of standardization in terms of AIoT proof-of-work proposed in the public chain only Hinder the mass adaptation which affects blockchain performance Guest fees are required for individual testing protocols	Proposed enhanced consensus only for public chain No hyperledger technology used
Fundamentals of blockchain and chain codes for design an efficient system [15]	2022	The main contributions are as follows: Differentiate between present and formal consensus methods and related updates Applicational examples of formal methods and their protocols are presented Used of PBFT-built trust in a blockchain-enabled solution is highlighted	High cost of computational energy is one of the emerged problems in this research The formal method consumes more network bandwidth for transactional validation	Scope of data preservation and management Multithreading limitations Fixed block size transactions are allowed
Comparison of public, private, and consortium consensus mechanisms [16]	2022	A new paradigm in the consensus to create an agreement among nodes that decide whether the block appends to form a new block for an efficient transaction executions To separate consensus into subdomains, such as proof-based mechanisms and voting-based schemes	Cost of data preservation Collaboration of two or more consensus together Evaluation metrics depend on the policy of fixed block size	Middleware BPFT Consensus optimization Proof-based and vote-based categories derived
A consensus for permanent public permissionless verifiable remote accessing for IoT [17]	2022	A novel remote attestation protocol proposed for public chain It presented a trustworthiness environment using blockchain, where entire historical attestation logs preserved A Proof-of-Concept implemented	Lack of security in the current structure of remote attestation Only accessible in the public permissionless blockchain network Required more network bandwidth for events of node transactions executions	Consume more computation energy for proven request Hyperledger Sawtooth technology used for testing purpose Not practice applied in large network
PBFT: An efficient performance for public chain transactions deliverance using multithreading mechanism [18]	2022	An optimized Byzantine Fault Tolerance (OPBFT) is proposed Compared to the proposed and previous PBFT mechanism Create infrastructure for Business to Business (B2B) for smooth transactions and deliverance	Crash-Fault Tolerant (CFT) system related limitations Scalability challenges Interoperability issues	Only applied for public permissionless structure Consumes less throughput and rate of latency when transmitting Cost of data preservation is high
Efficient method to enhancing blockchain PBFT consensus capability using weighted mechanism [19]	2022	A Weighted Byzantine Fault Tolerance (WBFT) consensus is proposed Three stages of communication are designed for efficient data transmission	Performance bottleneck: low throughput, high delay, sustainable related issues, targeted attacks, vulnerability, and performance instability	Influence malicious nodes reduces Probability of malicious behaviour reduces Increasing efficiency of consortium chain transactions

ledger system:

- Elapse time management
- Transaction optimization
- Multithreading computation
- Chronological chain organization

2.1. Blockchain and multithreading technology

The process to ensure all the nodes in the private chain network works properly as getting multithreading hierarchical executions and applying the same set of data integrity and privacy while transmitting updated logs among the participants is critical. It is only possible when technology collaborates with heterogenetic node connectivity using multithreading methods. However, the role of multithreading in blockchain provides promising solutions for various problems, such as single point of failure, data forgery, tampering, manipulation/alter information storage, and one of the most prominent prospects is protection against malicious attacks [14]. For instance, the development of blockchain solutions in SMEs creates different paradigms in terms of standardization, regulatory and compliance, cost metrics, and the scope of data management.

There are a few key terminologies that need evaluation while designing efficient blockchain-based Distributed Applications (DApps):

- Latency
- Throughput
- Privacy and Security
- Scalability
- Reduce delay

With the analysis of these mentioned assumptions, in this section, we find and separate different proposed methods, which are completely related to the proposed contributions that use multi-threading features for system efficiency and robustness. Along with that, we highlight research gaps that need to be investigated for futuristic developments, as discussed in Table 1.

3. Preliminary knowledge

In this section, the fundamental description of a middleware lightweight Proof-of-Elapsed Time (PoET) consensus and related problems in the private blokchcian network are mentioned. However, in this manner, a few basic preliminaries of the previous methods are discussed, along with that, the study formulates the lightweight PoET which are as follows:

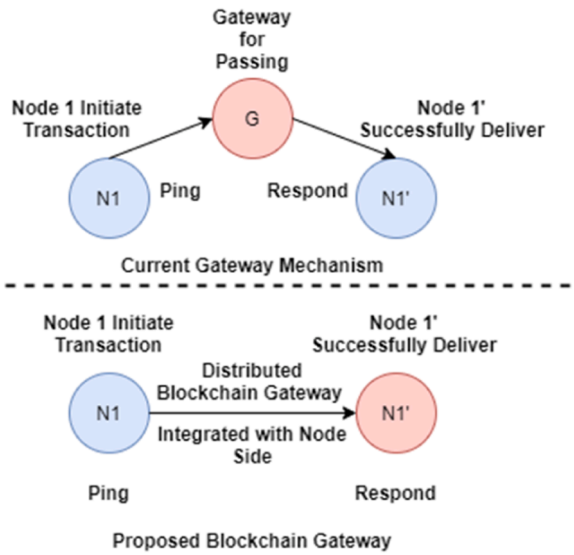

Fig. 2. Role of Blockchain Gateway in the Proposed B-LPoET.

3.1. Problem description and notations

The significant progress in blockchain consensus makes the system more reliable in terms of interoperable chain transmission and their related connectivity. However, on the other side, there have been made some differences that create limitations in the public, private, and consortium chains, where the legacy system cannot be communicated seamlessly yet [20,21]. For instance, the current standard of permissioned chain interoperability cannot allow the seamless transfer, including data and value across legal jurisdictions.

Table 2
Algorithm 1: Pseudo-Development of Chaincode and Consensus.

Major Assumptions to Initiate Features of Smart Contracts/Chaincode:
SMEs manager is responsible to handle transactions request throughout the process
Manager is authorized to add new SME in the chain
Schedule transactional requests
Check status of request verification and validation
Update ledger after transaction occur and exchange
Input Data: File[new]:X.[a.txt] -> int main()
File[new]->preserve (cloud-enabled DLT)
File[read]->preserve (cloud-enabled DLT)
File[write]->preserve (cloud-enabled DLT)
Declaration and Initialization:
SME registration,
SMER();
SME stakeholder,
SMES();
industrial transaction,
IT();
manufacturing transaction,
MT();
production transaction,
PT();
on-chain channel,
OCC();
off-chain channel,
OfCC();
new logs records,
NLR();
update logs records,
ULR();
cloud-enabled DLT,
CDLT();
update exchange,
UE();
transaction block size,
TBS();
Blockchain timestamp,
[execute];
Blockchain SMEs manager responsible to schedule, handle, organize, and manage ledger;
Expert responsible for ledger preservation;
Working Operations Processes:
if SMER() is != true
verify and validate request of joining;
then, add new SME in the SMER(),
and, exchange logs;
if IT(), MT(), and PT() == to initiate transaction and schedule,
then, each logs of executions are recorded = NLR();
and ULR() updates and exchange among the participating stakeholders;
also, all related activities are recorded such as SMER(), SMES(), IT(), MT(), PT(), OCC(), OfCC(), NLR(), ULR(), CDLT(), UE();
else verify request, update log, change state, preserve, and exchange;
stop executions;
else verify request, update log, change state, preserve, and exchange;
stop executions;
Consensus Queries: Lightweight PoET() Rules;
For transaction block == TBS(4MBs);
Transaction channels == OCC(implicit) and OfCC(explicit);
Digital signature == 51 % votes are required;
Blockchain gateways == decentralized interoperable private network;
Order of Nodes == Chronological;
Block Hash == NuCypher Threshold *Re*-Encryption;
Throughput == 4.33 % over 4–9 transactions;
Delay == 7.93 % over 4–9 transactions;
Outputs: SMER(); NLR();ULR(); and CDLT();

In the domain of SMEs, the hampering enterprise adoption of blockchain is another type of issue that raises. In this manner, there is a need for creating solutions capable of complying with standardized architectures and compliances [21].

Though it is necessary to design a blockchain gateway, the existing infrastructure of Internet routing gateways is capable of helping sequentially to enable decentralized interoperable private networks that initiate node-to-node communications. However, the designed blockchain getaway permits digital transactions and virtual assets to be transferred seamlessly between nodes via DApp, as shown in Fig. 2.

Within the Internet Engineering Task Force protocol, it provides an asset transfer platform that operates among the devices. There is ongoing work running on the current Internet Engineering Task Force protocol hierarchy for secure communication between two gateway devices. And so, it is highly needed the Open Digital Asset Protocol, which aims to provide cross-border communications. The benefit of designing this protocol is that it can be capable of handling a number of digital information interoperable transactions by leveraging blockchain gateway. For instance, the design of this gateway accepts the information to be shared via an information profile, for example, in accordance with regulatory compliance or policies. The transfer of information among the private chains through the gateway is equal to the swap and preserve mechanism that locks information in a blockchain. To guarantee a secure and fair sharing of information across gateways, the system required a computational energy minimum of 1 unit for 2–5 transactions/node.

To assure the properties of private chain consensus, the system needs to be enabled a secure and fair share of information. In this manner, the blockchain gateway integrates with a crash-recover mechanism that is associated with a core design key of the proposed blockchain gateway. Thus, a recovery protocol allied to a crash mechanism guarantees the source, where from it is initiated, and the targeted deliverance of blockchain distributed ledger technology which is updated consistently. For instance, the information of the transactions is taken from the source (node number, chronological order sequence, and chain where it is represented) and is persisted into the recipient distributed ledger. In addition, there is no duplication logs/transaction spending that can occur throughout the process.

However, we present PoET (a hyperledger sawtooth) consensus, a middleware for private chains of SMEs that focuses on providing a fair chance among the participating nodes in the network to schedule blocks of data transactions, such as industrial, manufacturing, and production. It associates with two different protocols with a designed blockchain gateway to improve the performance in terms of transaction redundancy, secure information sharing, resource consumption, and logs management. The main component of PoET connects with the extension of Open Digital Asset Protocol inspire into two different subphases, including commit and endorse channels. In this function, the system leverages a log preservation application programming interface that persists details on information/transaction transfer processes through a gateway to provide post-accountability and dispute-related solutions. However, the Lightweight PoET architecture is designed based on a layered hierarchy, allowing interoperable private network transactions using smart contracts/chain codes (as mentioned in Table 2) to be executed via blockchain gateway. The questions addressed in this paper are discussed as follows:

- Q1: What is the rate of reliability to execute transaction blocks in a minimum amount of time?
- Q2: What is the trade-off between the scheduled transaction of SMEs and the response?
- Q3: How the system increases the middleware performance in multithreading while reducing the rate of resource consumption, such as network bandwidth, computational power, and storage?
- Q4: How the system calls lightweight? And so, what to expect in the scope of data protection throughout?

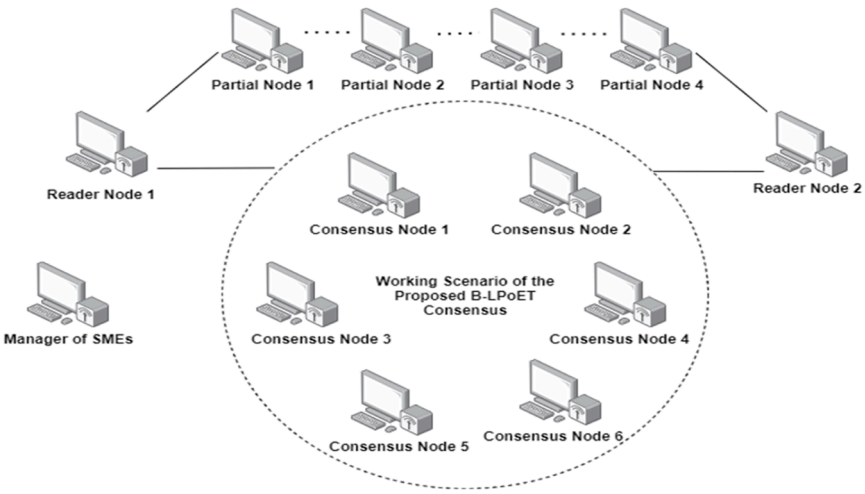

Fig. 3. Working Scenario of the Proposed B-LPoET After Middleware Multithreading Mechanism Involvement.

3.2. Problem formulation

For the lightning cost of current PoET, this paper contributes to three folds which are discussed as follows:

i. spreading the odd number of nodes of winning equitable through the big SMEs network members,
ii. each node in the network waits a random amount of time while going to sleep, and
iii. one with the shortest waiting time remains are awakened and contributed through a new block

However, creating trust between node transactions towards deliverance, the PoET integrates with Intel SGX control mechanism to form: (i) a certificate and (ii) trusted code. The designed certificate allows the PoET code to be regarded as not dependable for any specialized hardware. It is run over the same system infrastructure. On the other side, the trusted code is executed in a secure manner. In this manner, the system is capable to reserved and protecting the preservation environment, which cannot be forged or tampered with by the rest of the DApp. By this enhancement, we enable SMEs manager to prove other stakeholders for the purpose of executing trustworthy and efficient network code. It can improve privacy protection among the stakeholders while scheduling events of node transactions. In fact, the system makes certain clauses where a malicious cannot alter PoET trusted code after the secure setting, as shown in Fig. 3.

Whereas the node communication time throughout the process is

$$N_T = (n-1) * 2; \text{wheren} = \text{numberofnodesand} N_T = \text{Timetakesbythenodes};$$

A number of SME nodes entries SMEN = $\{N'_1, N'_2, N'_3, \ldots, N'_n\}$, such that the total entries have a total number of orders that need to be scheduled randomly, given by the time of its participation. Whereas the new blocks of SMEs industrial, manufacturing, and production transactions SMEB = $\{B'_1, B'_2, B'_3, \ldots, B'_n\}$, are assigned to the node such that who wins the timer object 'T_o', which is in between 1 and 3 units. An individual node in the designed private permissioned network is assigned a stopwatch that calculates waiting time and keeps up to note by the PoET controller. It also helps in the verification when the node sleeps/awakens. In addition, if the PoET controller authorizes the newly generated block whenever nodes awake, otherwise it is discarded. The selection and generation process for choosing a waiting is discussed as follows:

Process of Selection: In the process of creating a new block in the private permissioned network, individual joining nodes first share their generated certificate. After proper verification (using chain code), the node qualifies to receive a timer object.

Process of Generation: This process helps when a node's timer object gets expires while it awakens. The active node creates the hash/Cypher of each transactional block and submits it for permission to schedule transactions using NuCypher Threshold *Re*-Encryption (N→N'). It reduces the cost of additional computation of hashes throughout the process, which leads to effective proof-of-work (PoW) systems executions and management.

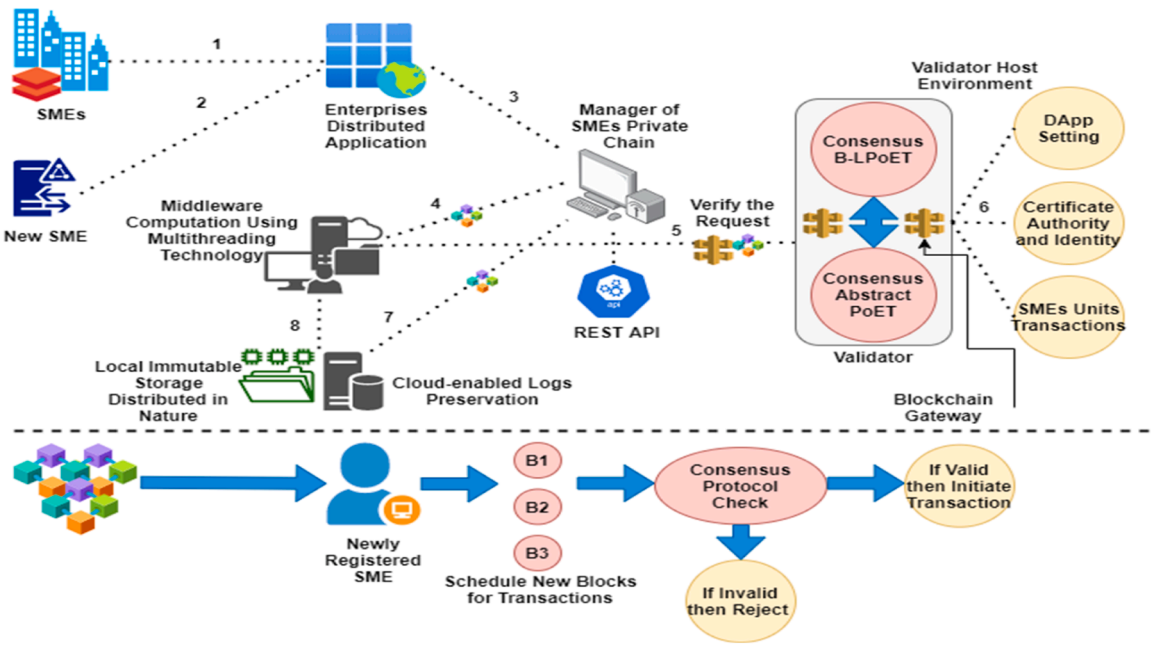

Fig. 4. Proposed B-LPoET Architecture.

For middleware Lightweight, a blockchain gateway is designed that is capable to interacts with private chains for the sake of issuing transactions, reading/writing states, etc. And so, the system can be able to perform computation based on the N'→B' interactions while the node awakens. However, there is an assumption, that a primary gateway integrates with the backup gateway for block-based secure transactional movement. For instance, the information transfer from one end to the other, data pushing/popping on the blocks, and scheduling blocks randomly over the private chain, as shown in Fig. 3.

The rule of thumb for reducing resource consumption throughout the process of transaction scheduling toward execution is highlighted as follows:

- heterogeneous connectivity between nodes
- local-area network is designed with 4MBps
- the block size for the individual transaction is minimized up to 4 MB
- bandwidth for a single transaction per execution is 1MBps
- redundancy check using a block duplication mechanism
- executed logs are preserved in cloud-enabled distributed storage/local distributed storage of SMEs

4. Proposed B-LPoET architecture

Fig. 4 presents the working operations of the proposed B-LPoET, a blockchain-enabled middleware lightweight Proof-of-Elapsed Time (PoET) for efficient distributed transaction execution and security using a multithreading mechanism. This proposed B-LPoET architecture is divided into eight (8) different sub-portions. However, in the first step, the number of SME registered stakeholders initiate transactions over the designed B-LPoET permissioned private chain via Enterprises Distributed Application (DApp). For instance, if a new SME wants to participate in this chain (who is not enrolled previously), then it follows the instruction when opening the same DApp interface. After careful submission of the enrolment request, the manager of SMEs plays a vital role to identify certification assigned by the competent authority (regulatory authority/ministry of the government) before allowing new SMEs to join the chain. The manager is also responsible to verify the request, validating the request, ledger maintenance, and even monitoring throughout. At the same time, REST API reset all occurred transactional requests, such as new SME requests, registered SMEs transaction initiation requests, and probably all the logs which get entertained, simultaneously.

Based on this proposed B-LPoET architecture, we design, create, and deploy SME distributed application (DApp) associated with chaincode. There are four different functions introduced, which aim to automate the request of access, authentication, authorization, transaction initialization, log management, and preservation. Meanwhile, after getting requests executions from the manager, the middleware system computes individually. Its sent execution verification through the blockchain gateway, where the design B-LPoET consensus validates trusted code after analysis DApp setting, certificate identity, and category of SMEs units. After proper investigation, it sends back validation results to the manager for further decision-making. This procedure enhances resource utilization as compared to the previous node-to-node and machine-to-machine-enabled computation. However, the logs are generated whenever a single transaction fluctuation occurs. In this manner, the cloud-enabled logs preservation mechanism is deployed, which is immutable in nature and based on a locally distributed storage hierarchy for cost reduction. It can handle off-chain (all explicit transactions occurring, as shown in Fig. 4(7)) and on-chain (all implicit transactions occurring, as shown in Fig. 4(8)) transactions.

On the other side, Fig. 5 shows the working hierarchy of the newly registered SME in the designed blockchain private permissioned network, B-LPoET chain. After getting authorization, the registered SME schedules transactions in the form of blocks (each with a 4 MB fixed block size), which are aligned in chronological order. Each block verifies before moving towards executions with the B-LPoET consensus check, where if the block transaction is valid then it gets executions otherwise rejected.

4.1. Pseudo-Development of chaincodes

In this context, the pseudo-implementation of chaincode is presented, along with the working operations and procedure of executions. With the objective of deploying verify and validating the request of individual transactions of SMEs. Some major assumptions need to follow for the sake to initiate the functions of the designed chaincode, which are as follows: (i) the SMEs manager is responsible to handle requests generated by individuals, (ii) the manager allows to access the ledger, (iii) monitor each request and related schedules, (iv) check status of requests verification, and (v) update ledger and exchange details. In the chaincode initialization, the function SMER() performs a key role to add a new registry when receiving requests from newly open SMEs in the chain. The manager receives the request and verifies it once, then the chaincode auto-validates the list of nodes and allows new SMEs to initiate

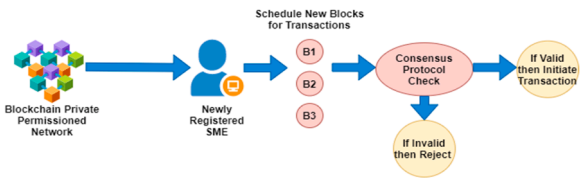

Fig. 5. Working Hierarchy of Newly Registered SME in the B-LPoET Private Chain.

transactions. On the other side, the function NLR() concurrently updates the changes logs occurring throughout the addition (new transactions) in the chain and preserves it in the cloud-enabled distributed local storage with the help of the ULR() function. The main objective is to maintain a distributed ledger immutably for futuristic investigations.

For transactions management, the function CDLT() performs an important role to capture update logs, examining individuals, analyzing duplication, storing changes, and maintaining change reports. For instance, the updated logs are then available for exchange among the participating stakeholder in the proposed private permissioned blockchain. The consensus changes (Lightweight-PoET() rules) and the digital signature mechanism that is associated with this chaincode are highlighted as follows (as mentioned in Table 2):

- For transaction block = TBS (4 MB)
- Transaction channels = OCC (implicit) and OfCC (explicit)
- Digital signature = 51 % votes are required to revert transaction
- Blockchain gateways = decentralized interoperable private network
- Order of Nodes = Chronological order
- Block Hash = NuCypher Threshold Proxy *Re*-Encryption
- Throughput = 4.33 % per 4–9 transactions
- Delay = 7.93 % per 4–9 transactions

5. Results and discussion

This section discusses the superiority of the proposed middleware B-LPoET, a private permissioned chain based on multithreading technology. For instance, the evaluation of the B-LPoET is divided into seven different folds. First, the average relationship between the number of transactions over time(s), where the highest value number of transactions touches 266 while the fluctuate time is 4000 s (such that, the result is 266/4000 = 0.0665) in the complete process of examination, as shown in Fig. 6.

However, Fig. 7 illustrates the evaluation of the average transaction commit rate, whose metrics are the average transaction and the transactions per/s. The experimental results highlight that the average transaction metric touches 266 transactions while the rate of transaction execution per second is 4000 s (such that the result is 266/4000 = 0.0665). On the other end, the simulation is performed for examining the average relationship between the average transaction commit rate and consensus latency. The evaluation results show that the total transaction committed by the proposed B-LPoET is 271, where the consensus latency is noted as 8000(ms). It is well noted that the average transaction commit rate is 271/8000 = 0.033875, as shown in Fig. 8.

The evaluation metrics show that it consumes a low latency while supporting more nodes up to 4–9/transactions (as highlighted in Table 3), effectively decreasing the delay in consensus executions by reducing intercommunication load between nodes down to 4.33 % and decreasing the usage of computational power down to 7.93 %, respectively.

Fig. 9 illustrates the evaluation of the average relationship between the number of nodes scheduled for transaction execution, whose metrics are the number of scheduling executions and the consensus latency. The experimental results highlight that the average transaction metric touches 271 transactions while the rate of transaction execution per second is 8000 s (such that the result is 266/4000 = 0.033875).

However, the simulation is performed for examining the average relationship between the number of nodes scheduled and the node communication time(s). The evaluation results show that the total number of nodes scheduled by the proposed B-LPoET is 237, where the node communication time is noted as 7000(s). It is well noted that the average transaction commit rate is 237/7000 = 0.033857, as shown in Fig. 10.

And so, another simulation evaluation noted that the average relationship between the number of transactions over time(s), where the highest value of the number of nodes scheduled touches 237 while the fluctuating node communication time(s) in 7000 s (such that, the result is 237/7000 = 0.033857) in the complete process of examination, as shown in Fig. 11.

In the end, the simulation of the proposed B-LPoET performs for finding the average relationship between the number of node-based transaction executions and the average waiting time(s). The evaluation metrics show that the total nodes from the rest (0) to

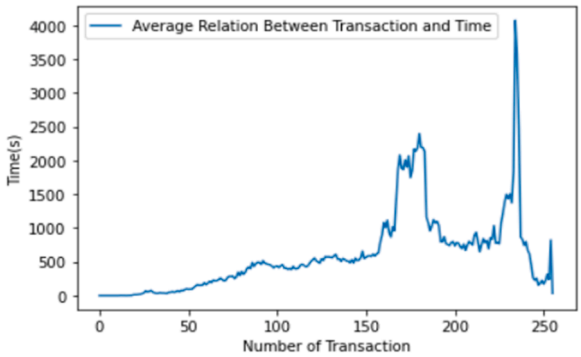

Fig. 6. Average Relation Between Number of Transactions and Time(s).

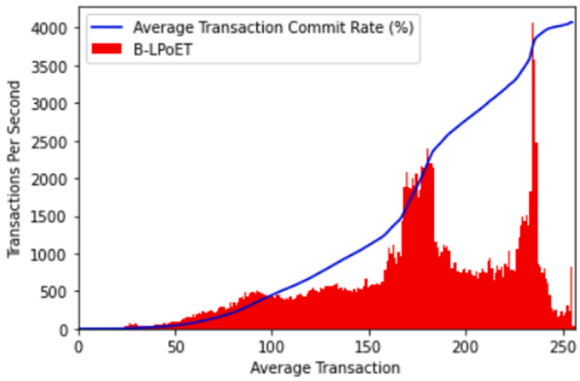

Fig. 7. Average Relationship Between Transaction Commit Rate Over Time(s).

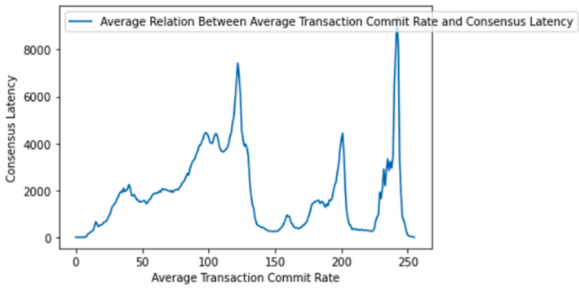

Fig. 8. Average Relationship Between Average Transaction Commit Rate and Consensus Latency.

Table 3
Evaluation Metrics of the Proposed B-LPoET.

Latency Test (1)	Delay Test (1)	Network Resource Test (1)	Cost of Computation Test (1)	Latency Test (2)	Delay Test (2)	Network Resource Test (2)	Cost of Computation Test (2)	Average Evaluation (%)	Net Evaluation (%)
8000 ms	1 unit	Average	Average	4000 ms	1 unit	Low	Low	4.33 %	7.93 %

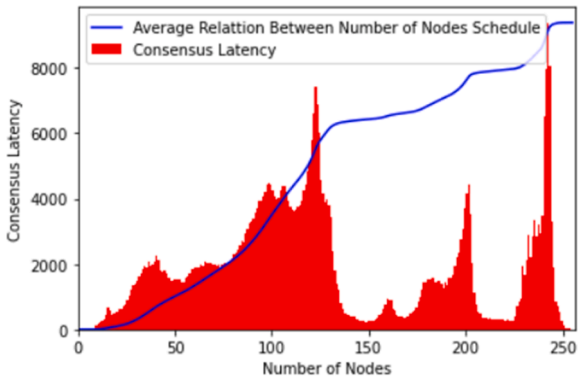

Fig. 9. Average Relationship Between the Number of Nodes Scheduling and Consensus Latency.

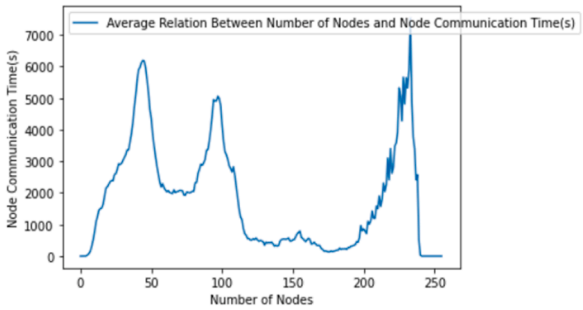

Fig. 10. The Average Relationship Between the Number of Nodes Schedule and Node Communication Time(s) Test (1).

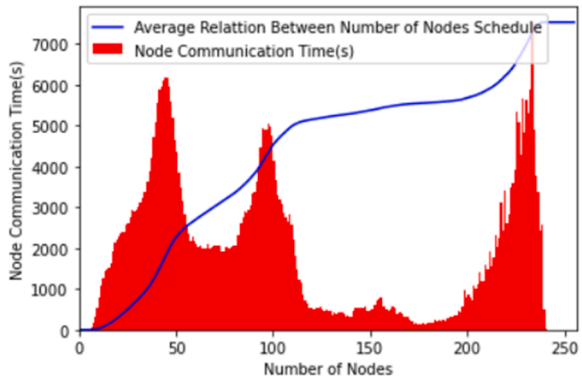

Fig. 11. The Average Relationship Between the Number of Nodes Schedule and Node Communication Time(s) Test (2).

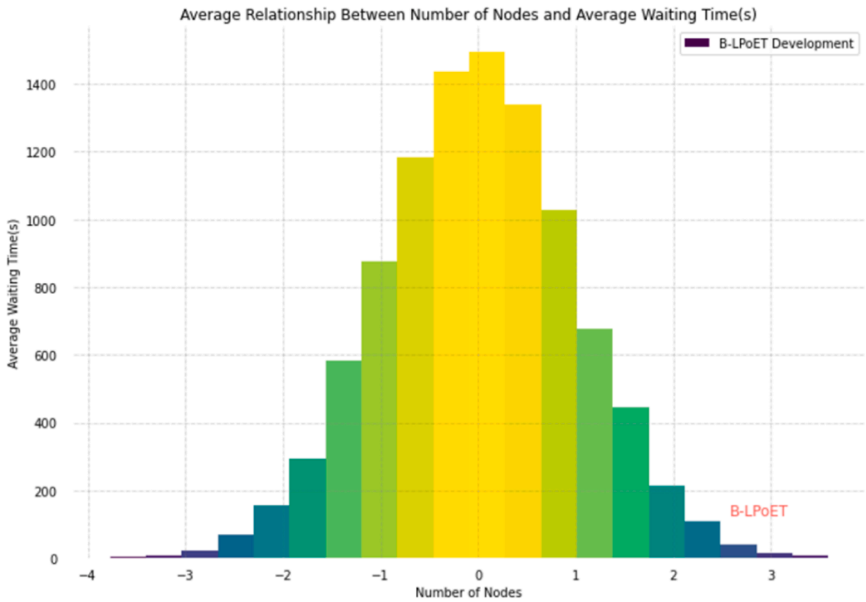

Fig. 12. The Average Relationship Between the Number of Nodes and the Average Waiting Time(s).

Table 4
Comparative Table for State-of-the-Art methods Comparision.

Previous Method (1) [9] Year of the Study: 2022	Previous Method (2) [24] Year of the Study: 2022	Previous Method (3) [25] Year of the Study: 2022	Proposed B-LPoET Year of the Study: 2023
The evaluation matric of the proposed B-LPoET with other state-of-the-art are mentioned as follows:			
Blockchain DLT: Public blockchain **Network:** Permissionless network **Node interconnectivity:** Sequential Block size: Variable **Encryption mechanism:** Hash (SHA-256) **Communication channels:** Single chain transmission **Consensus:** PBFT **Resource consumption:** High **Cost of privacy:** High **System's Accuracy:** Average	**Blockchain DLT:** Private blockchain **Network:** Permissioned network **Node interconnectivity:** Gazette structure **Block size:** Variable **Encryption mechanism:** Hash (SHA-256) **Communication channels:** Single chain transmission **Consensus:** FT blockchain transform mechanism **Resource consumption:** High **Cost of privacy:** High **System's Accuracy:** Average	**Blockchain DLT:** Consortium blockchain **Network:** Hybrid network **Node interconnectivity:** Bitcoin like nodes structure **Block size:** Variable **Encryption mechanism:** Hash (SHA-256) **Communication channels:** Single chain transmission **Consensus:** PBFT **Resource consumption:** High **Cost of privacy:** High **System's Accuracy:** Average	**Blockchain DLT:** Private blockchain **Network:** Permissioned network **Node interconnectivity:** Chronological order **Block size:** Fixed (4 MB) **Encryption mechanism:** NuCypher threshold proxy re-encryption **Communication channels:** Two channels (on-chain and off-chain) **Consensus:** Lightweight-PoET **Resource consumption:** 7.93 % (Low) **Cost of privacy:** 4.33 % (Low) **System's Accuracy:** High

3 nodes-initiated transactions through the blocks while the average waiting time for each node receive the 1400s (such that the number of nodes = 0–3 and the average waiting time = 1400s: 3/1400 = 0.002142), as shown in Fig. 12.

5.1. Comparison of the proposed B-LPoET with other State-of-the-Art methods

Table 4 presents a comparative section between the proposed B-LPoET, and other state-of-the-art methods based on the working operations, criteria of transactional requests, scheduling, management, efficiency, and resource utilization features. In fact, there are different methodologies, frameworks, architectures, approaches, and schemes also proposed that claim superiority in the scenes of their performances and evaluations. However, as the same, the author of "an efficient FT Blockchain consensus transform for IoT" [22] and the author of "a secure distributed blockchain PBFT consensus for smart IoV" [23] presented two distinct methods for different chains and highlighted their originality. First, describe the usage of FT in consortium blockchain to enhance the efficiency in terms of privacy and security in IoT-enabled information preservations. While the second discussed the role of consensus in public interoperable chains and the use of hyperledger technology.

5.2. Open limitations in the proposed B-LPoET

This section discusses the involving challenges, limitations, and issues while designed, developed, and deployed the B-LPoET DApp with lightweight PoET consensus mechanism in the private permissioned chain are as follows:

5.2.1. Outsource computation for middleware data management

The Internet of Things (IoT) has received vast adaptation in different domains, especially in academics and the industrial environment. Currently, various blockchain-enabling IoT solutions rely on partial centralized protected computations, which incurs high computation and requires more storage space as compared to the previous infrastructure, even if it leads single point of failure as well [26–28]. Thus, the security and privacy concerns in the blockchain-enabled IoT distributed applications make data exchanging and mining related to encryption data challenging problems [29]. Therefore, it is highly recommended to design and deploy a secure outsource computational environment for the purpose to reduce the cost of resource utilization and protection. However, the Internet of Everything (IoE: the extension of IoT with cloud computing) raises as one of the critical outsourcing factors that make data closer to runtime execution in the same machines. The experts in blockchain-enabling IoT need to integrate with this newly involving paradigm and propose better solutions for future developments.

5.2.2. Anonymous authentication in private vs consortium chain

For the anonymous authentication issue, Zero-Knowledge Proof Primitive (ZKPP) plays a vital role in terms of enabling a stakeholder (a person who wants to be data secret) who possesses information secret to prove to other stakeholders (a verifies who is only able to verify) [26,27]. Secret information satisfies a certain set of knowledge without sharing the actual information; that is called zero knowledge. In this manner, the ZKPP achieves anonymous authentication, privacy preservation exchange, asset transfer, and ledger management in the public, consortium, and private chains [28–31]. It also leverages zero-knowledge protocols to offer anonymous authentication for stakeholders in their transactions via DApp over a distributed network.

5.2.3. Role of digital signature

With the unique concept of digital signature provide an impossible tampering and forgery situation for malicious insiders. The proposed mechanism uses number theory, where the owner owns both the public and private keys as a pair form [29]. For instance, the

public key uses to encrypt owner transactions that guarantee the security and privacy of sensitive key information. On the other side, the private key is the only accessible factor to decrypt the encrypted information. However, a digital signature is a primary building block in blockchain DLT, mainly used to authenticate events of node transactions. Each and every node in the private chain network verifies submitted transactions and checks all participating nodes' work to agree on a proven state [29]. Although it increases individual node integrity, transparency, privacy, and security but consumes more cost in terms of computation, energy, and power. To lightning this, the system needs to update the current hierarchy by replacing the cryptographic mechanism with NuCypher technology.

5.2.4. Regulatory and compliance

The vast adaptation of blockchain private permissioned networks for SMEs' scalability makes DLT more effective and reliable in the recent environment [27,29]. However, the current protocols of joining and initialization with the government are one of the challenging factors [29]. The governmental authorities need to invest more time to design and deploy a new way for industrial modular infrastructure, where it can follow compliance rules to open more SMEs, layered-based transactions with no charges, services deliverance from cross-chains [30–33], etc. Meanwhile, the units of SMEs need to reconsider their process hierarchy when initiating transactions, including scheduling, examining, analyzing, managing, organizing, optimizing, executing, and storage-related implications as ascription. These two-sided limitations create lots of hierarchy; considering such issues, the federal ministry of SMEs and the local authority need to integrate various third-party enterprises to resolve the middle bridge around them. Design a novel and secure environment for information management and transmissions using blockchain-enabled DLT. By deploying this, the system reduces more costs in terms of resource utilization and usage of power, which will directly affect industrial production and revenue.

6. Conclusion and future direction

This paper discusses the current research gaps raised in the private permissioned chain environment and their proposed consensus mechanisms. While careful investigation, this paper separates middleware, lightweight and optimization, resource consumption, cost of data scope scheduling, management, and scalability-related problems that pose a serious concern. Thus, choosing all these gaps in a single domain is one of the critical tasks. However, this paper contributes to three different folds. First, this paper presents a modular infrastructure for novel and secure permissioned private chains. Second, a middleware lightweight PoET is proposed for efficient transaction executions using a multithreading strategy. Third, it addresses a few critical issues that involve in the previous systems, such as high computational cost, consuming more network bandwidth, decrease latency throughout, delay increment, and preservation related.

Therefore, in this paper, the mentioned limitations are tackled simultaneously, while developing and deploying the B-LPoET system infrastructure. The evaluation metrics of the B-LPoET show that it consumes a low latency while supporting more nodes up to 4–9/transactions, effectively decreasing the delay in consensus executions by reducing intercommunication load between nodes down to 4.33 % and decreasing the usage of computational power down to 7.93 %, respectively. In this manner, the role of chaincode is crucial; it automates the transactional requests (such as SME registration, new transaction, update, and record preservation) verification and validation processes. It creates stability in terms of scheduling requests, examining, analyzing, storing, reporting, and maintaining them in an immutable nature, such as the designed cloud-enabled distributed local storage for a low cost. After receiving such a great result, the proposed B-LPoET is considered a good candidate for real-time industrial adaptation.

Author contributions

- The author of this paper contributes equally.
- The authors agreed on the online/published version of this manuscript.

Declaration of competing interest

The author of this paper declares there is no conflict of interest.

Data availability

No data was used for the research described in the article.

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